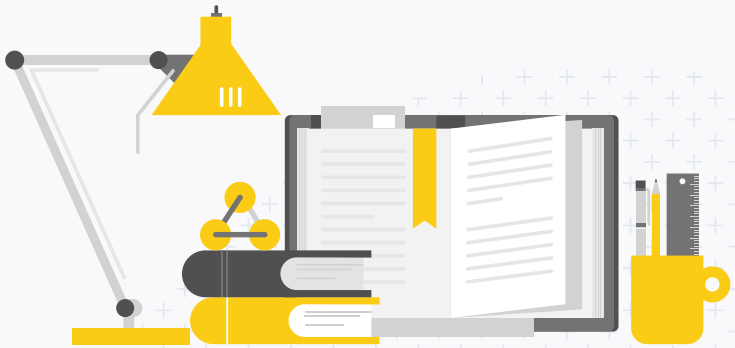
StudyTime

A Centralized Web-Based Database Platform for University Hall Seating, Library Lending, & Space Discipline



Course: Database Management Systems (Sessional)
Course No: CSE 252

Submitted By,
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Submitted To,
Sabiha Anan,
Assistant Professor
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Motivation & Real-World Hall Problems

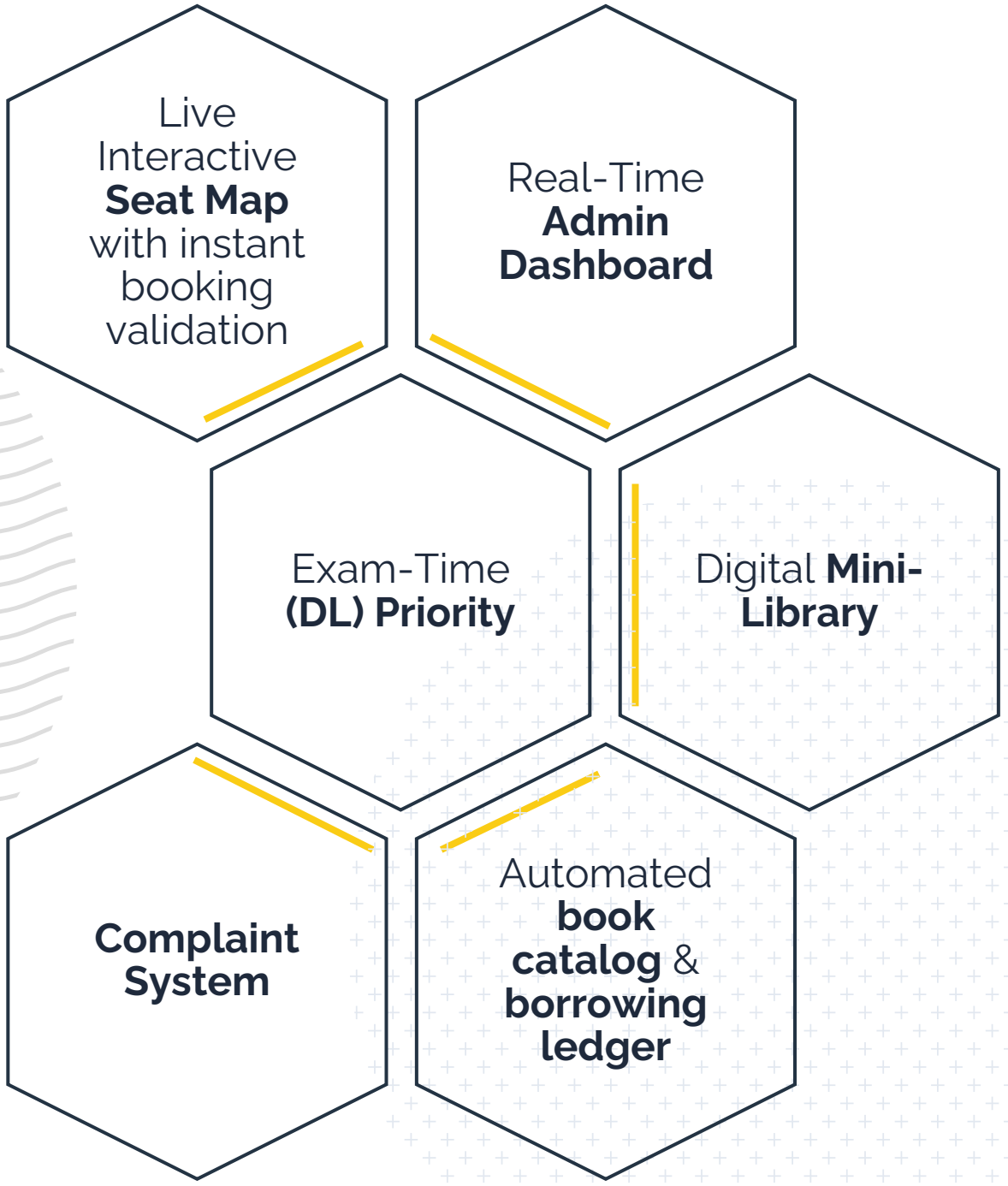


- **Phantom Seat Occupation**
  Belongings (books/bags) left on desks for hours without physical presence.
- **Seat Hoarding**
  Students occupying multiple seats during peak hours (e.g., DL time).
- **Unmanaged Mini-Libraries**
  Hall libraries remaining locked/unmanaged with no borrowing tracking.

Existing Solutions vs. StudyTime

Feature/System	Skedda	Google Sheets	LibCal	StudyTime
Context Customization	Corporate focus	Generic Spreadsheet	Formal Library Use	Native Hall Model
Real-Time Color Seat Map	Complex Interface	Not Available	Fixed Layout	Dynamic Live Floor Plan
DL Priority Allocation	Not Supported	Not Supported	Not Supported	Supported
Complaint Handling	Not Supported	Not Supported	Not Supported	Integrated System
Mini-Library Lending	Not Supported	Not Supported	Premium Add-on	Integrated Ledger

Key Features At A Glance

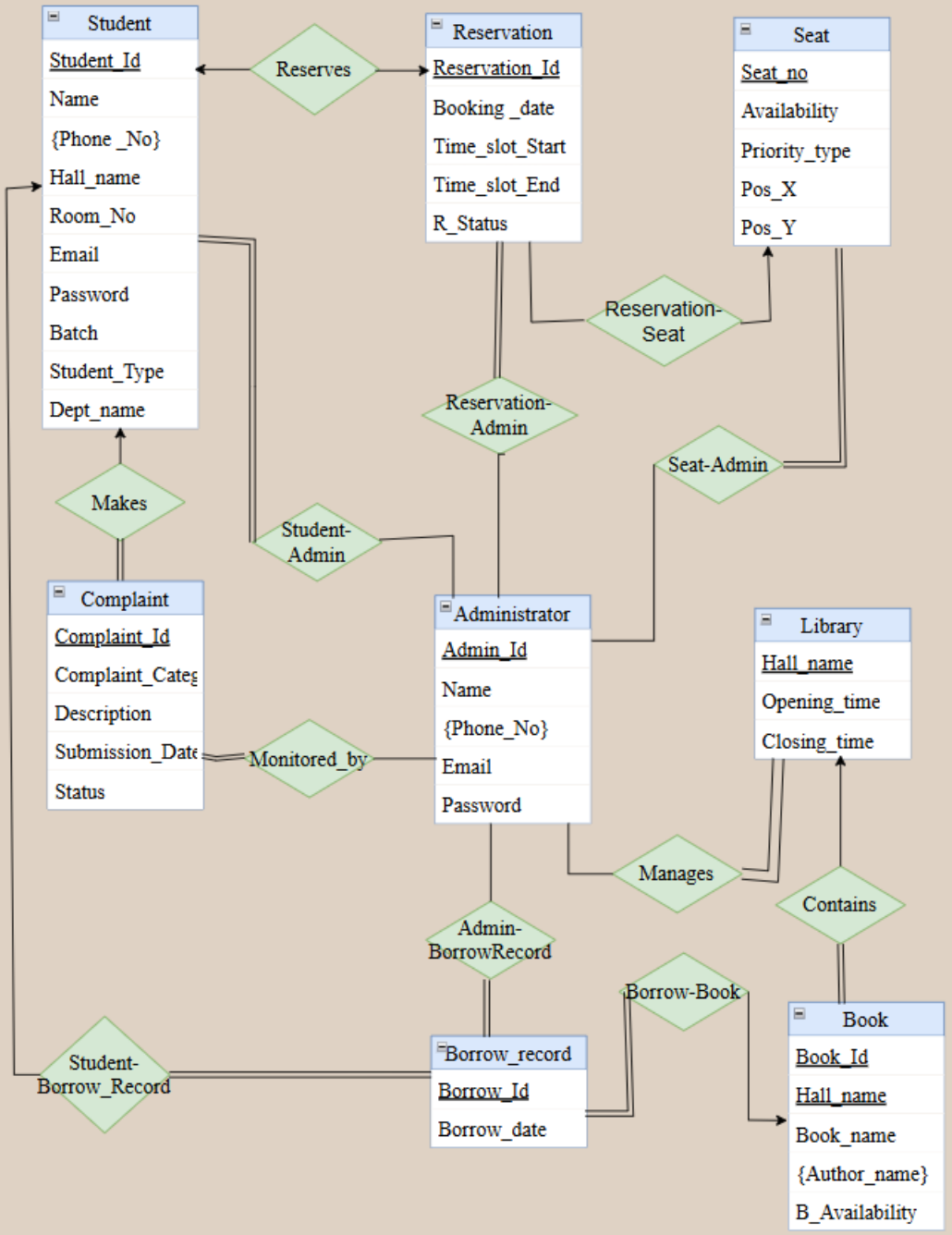


FINAL DATABASE DESIGN

Entity Relationship Diagram

Our Entities

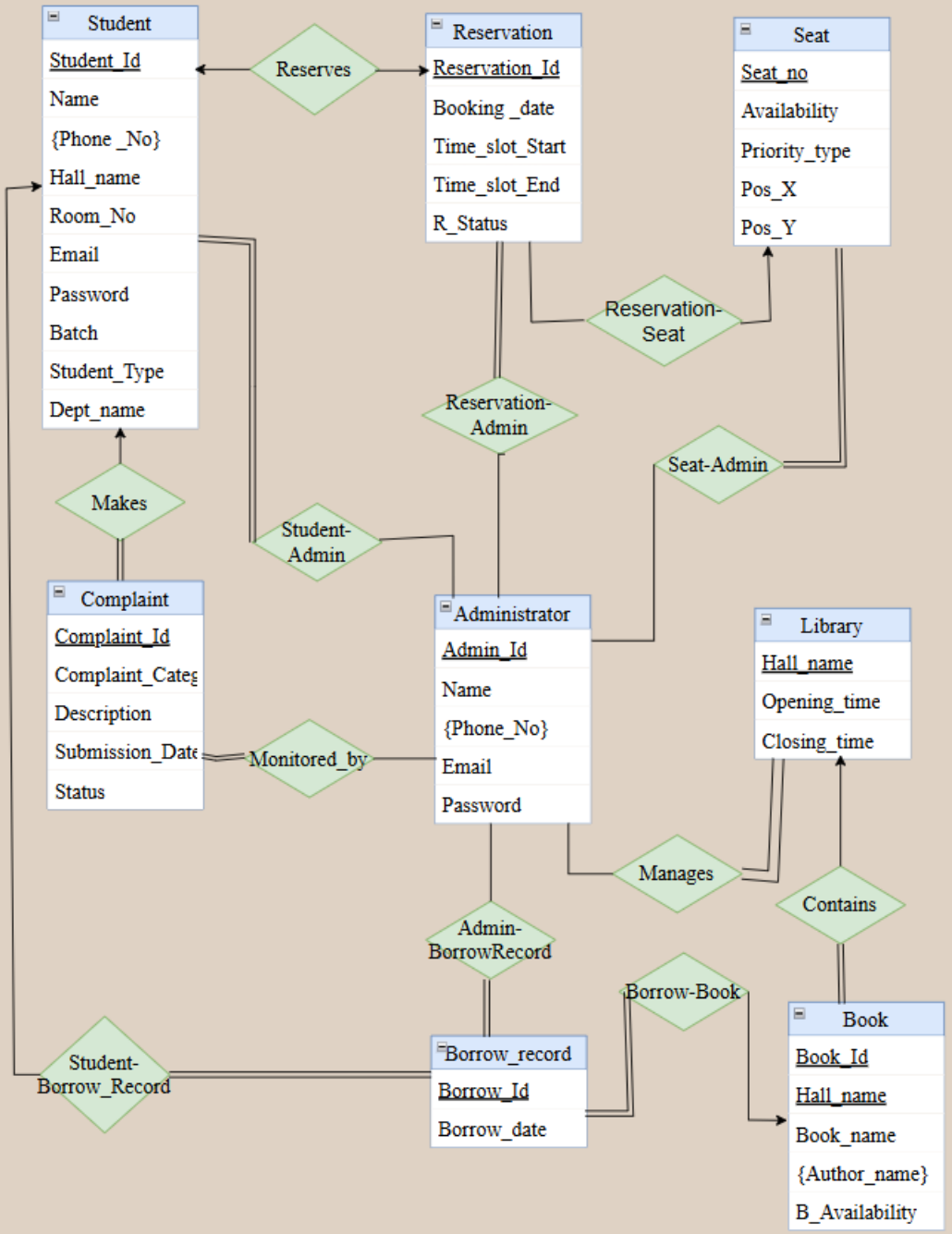
- Student
- Reservation
- Seat
- Complaint
- Administrator
- Library
- Borrow_record
- Book



Relational Mapping Creation

8 Entities generate 8 tables ,
each containing their
attributes as columns

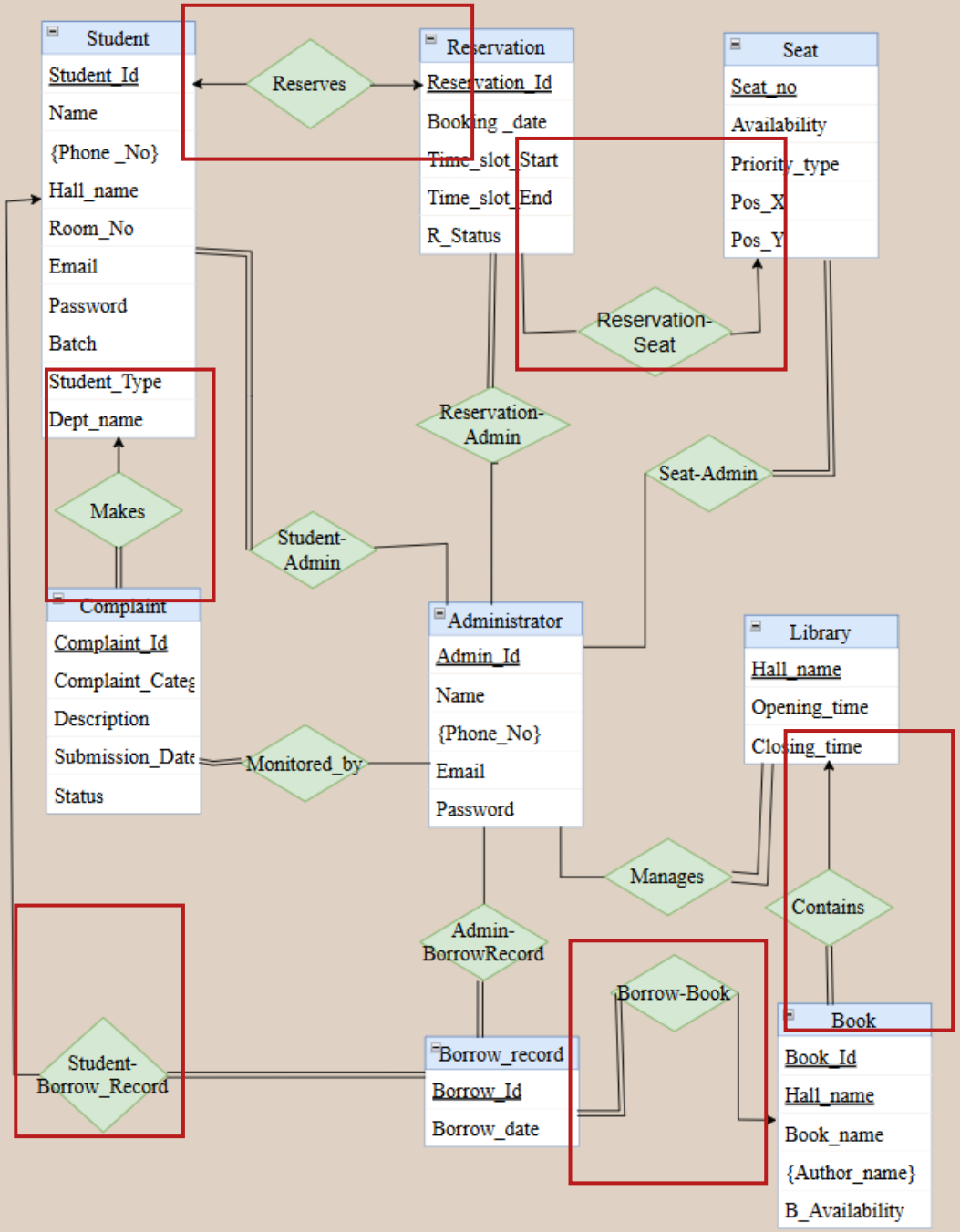
- Student
- Reservation
- Seat
- Complaint
- Administrator
- Library
- Borrow_record
- Book



Relational Mapping Creation

1 one-to-one relationship
Primary Key of any one table set to foreignkey of another

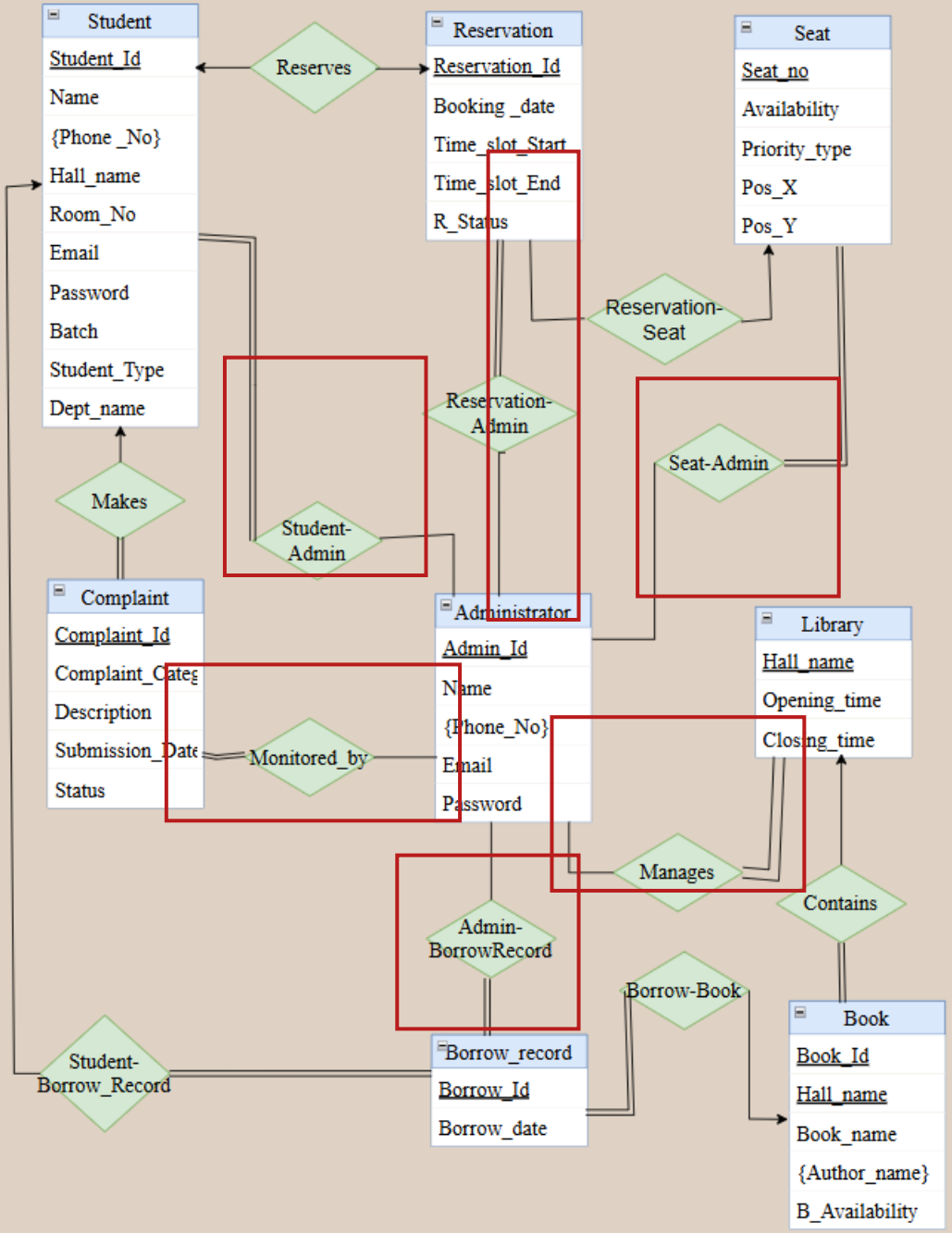
5 one-to-many relationship
Primary key of one side set as foreignkey of many side table



Relational Mapping Creation

6 many-to-many relationship

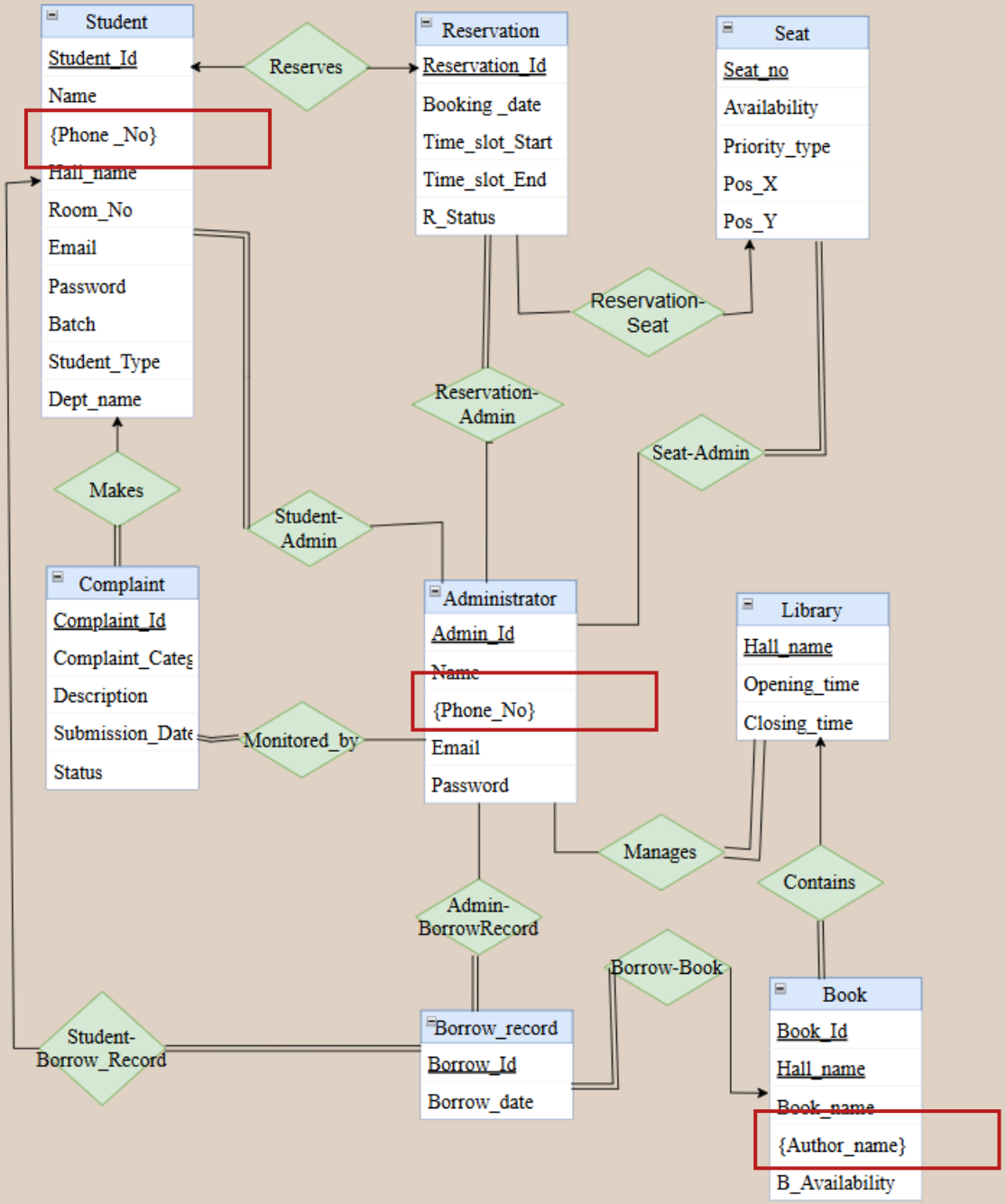
Separate Relational Table created with primary keys of both entities, both of which represent a composite primary key



Relational Mapping Creation

3 Multi-valued Attribute

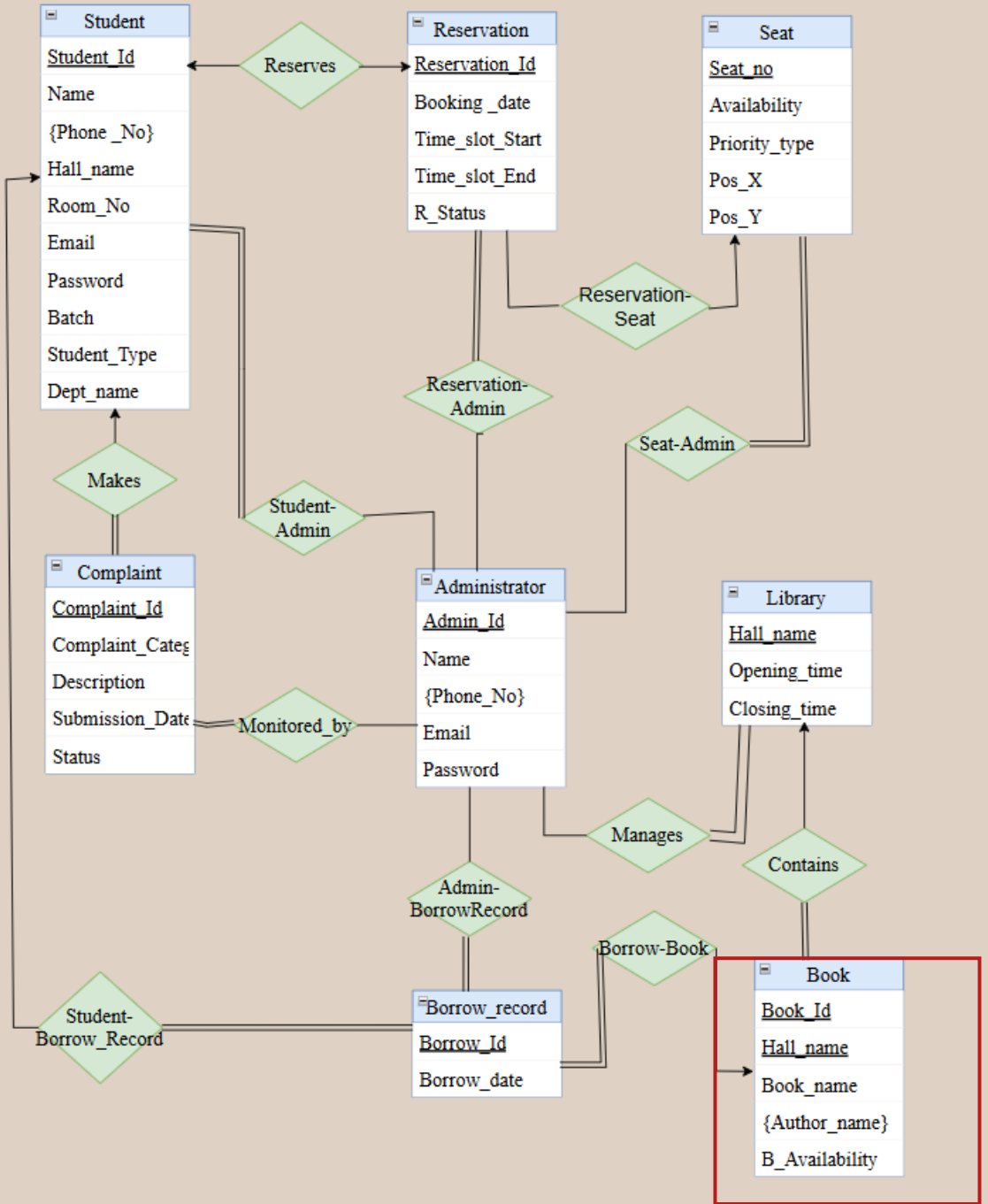
Constitute new tables, each containing the primary key and the attribute as a composite primary key



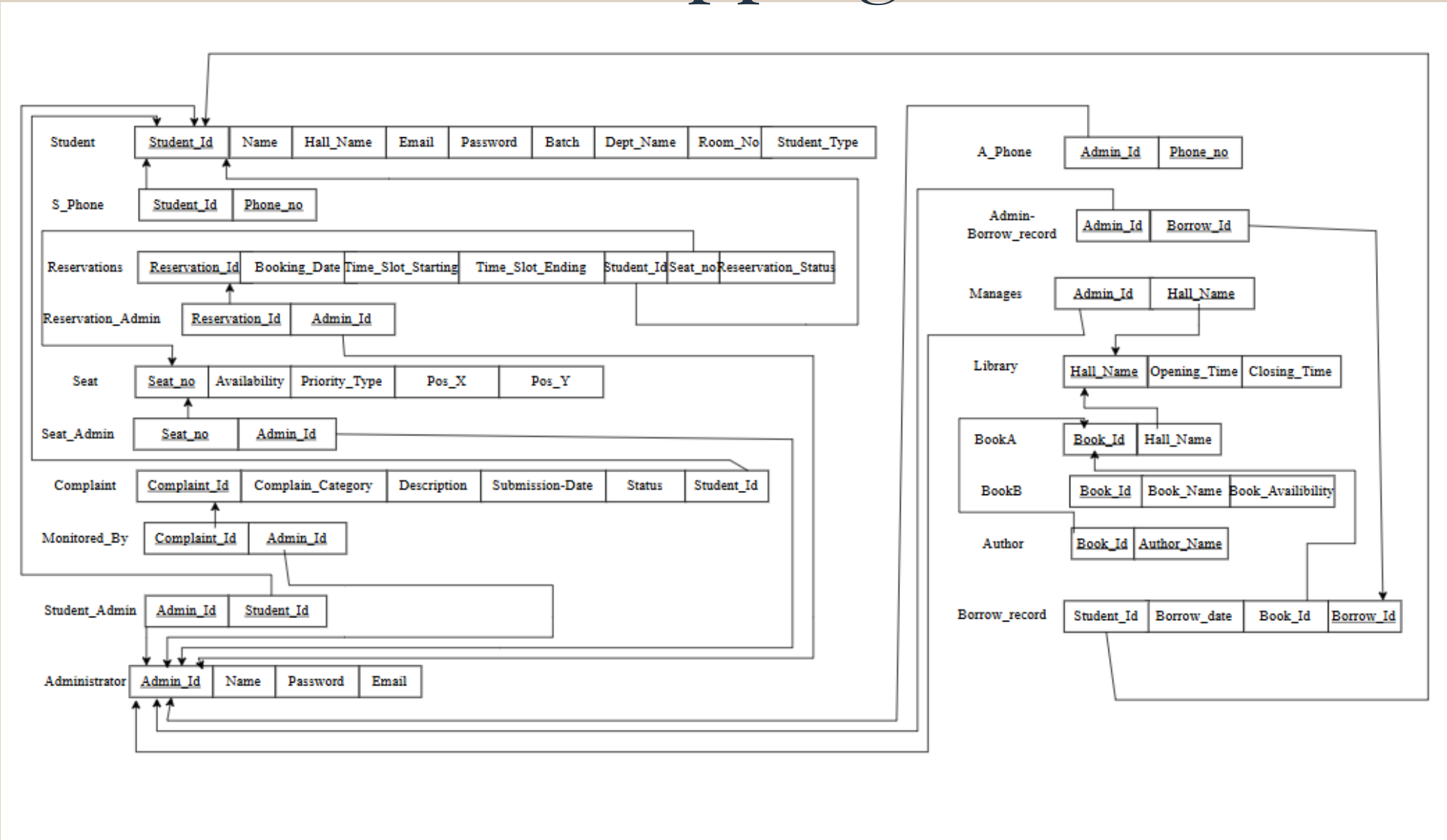
Relational Mapping Creation

Normalization

B_Availability depends on Hall_name. So, it violates 3rd Normal Form. So, Book Table is decomposed to BookA(Book_Id, Hall_Name) and BookB(Book_Id, Book_name, B_Availability)



Final Relational Mapping



Reservation Table

reservation_id	booking_date	reservation_status	time_slot_ending	time_slot_starting	seat_no	student_id
1	2026-08-01 16:21:53.781443	CANCELLED	2026-08-01 18:21:00.000000	2026-08-01 16:36:00.000000	A-01	1
3	2026-08-02 06:58:18.524935	CANCELLED	2026-08-02 08:57:00.000000	2026-08-02 07:12:00.000000	A-04	4
4	2026-08-02 06:59:06.239684	ACTIVE	2026-08-20 08:57:00.000000	2026-08-18 19:12:00.000000	A-04	4
5	2026-08-02 06:59:33.595792	ACTIVE	2026-08-02 08:59:00.000000	2026-08-02 07:14:00.000000	A-04	4

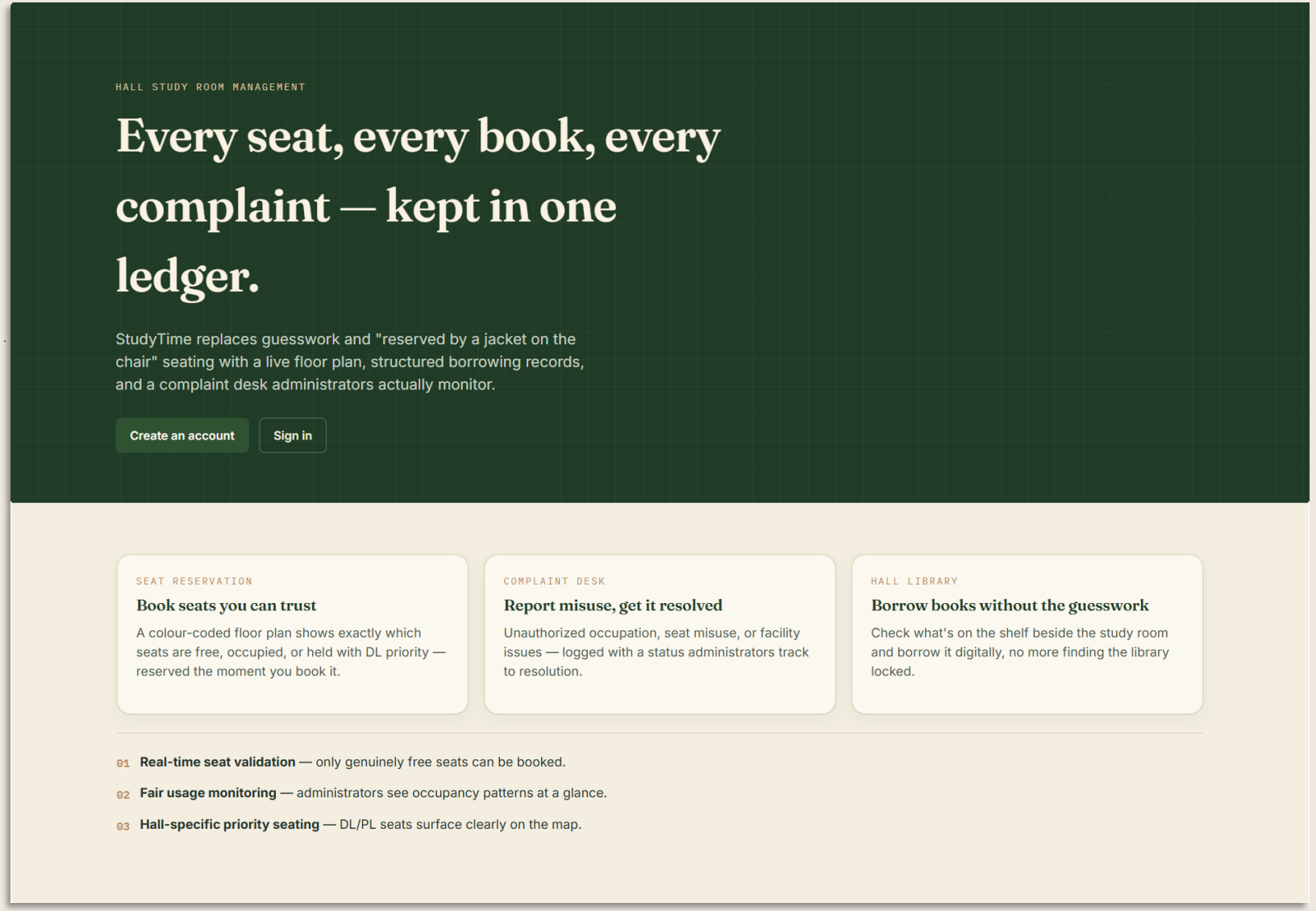
Seat Table

seat_no	availability	pos_x	pos_y	priority_type
A-01	1	236	155	General
A-02	1	236	192	General
A-03	1	237	226	General
A-04	1	362	150	General
A-05	1	360	186	General
A-06	1	362	222	General
B-01	1	396	150	General
B-02	1	395	185	General
B-03	1	394	221	General
B-04	1	497	147	General
B-05	1	498	183	General

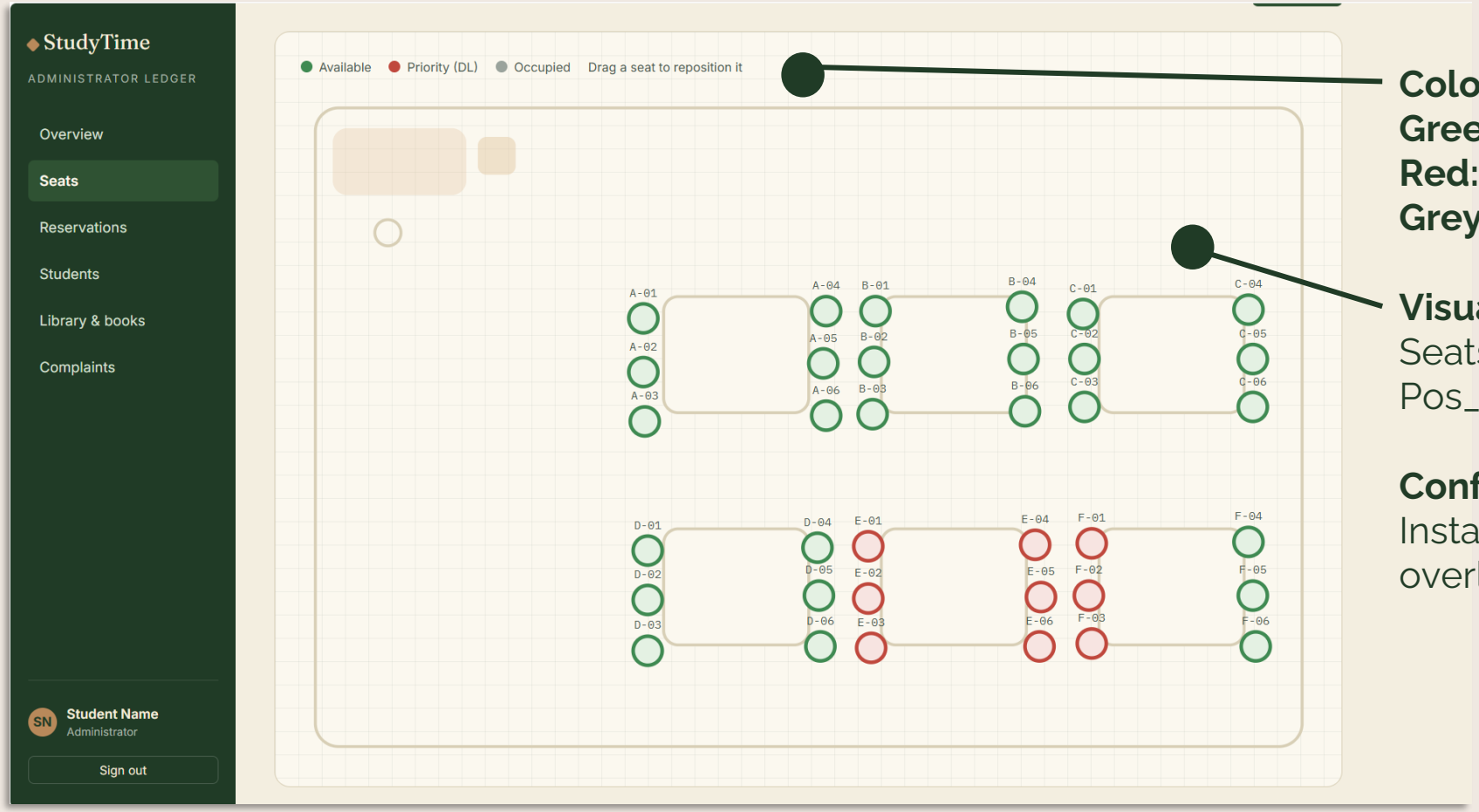
Each table is stored in our database as per the Relational Mapping

SYSTEM IMPLEMENTATION

Landing Page



Interactive Real-Time Seat Map & Booking

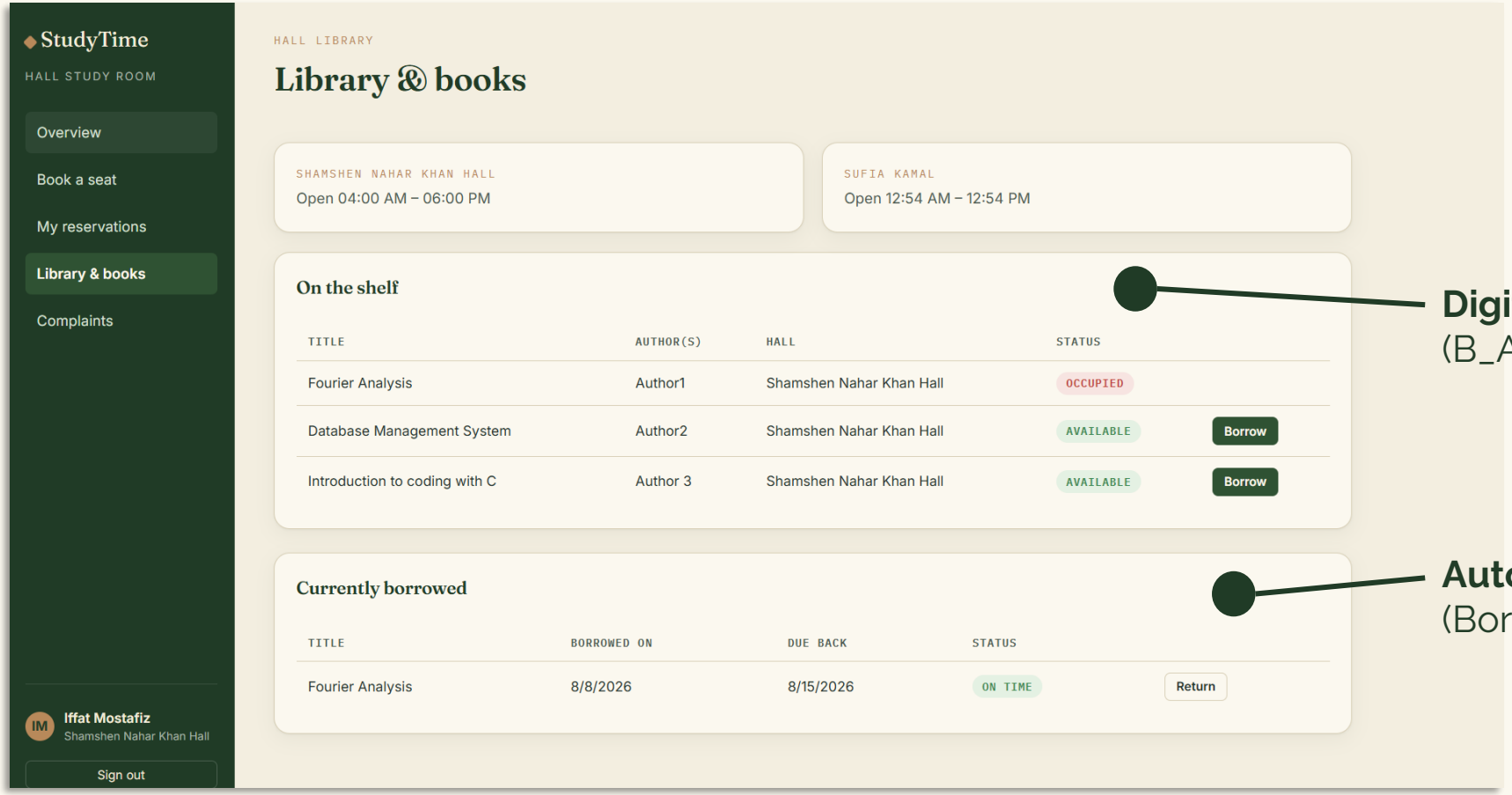


Color-Coded Seat Status
Green : Free / Available for instant booking.
Red: Reserved / DL Student Priority seating.
Grey: Occupied / Maintenance.

Visual Spatial Grid
Seats plotted via exact coordinates (Pos_X, Pos_Y)

Conflict Resolution
Instant server-side validation blocks overlapping time-slots.

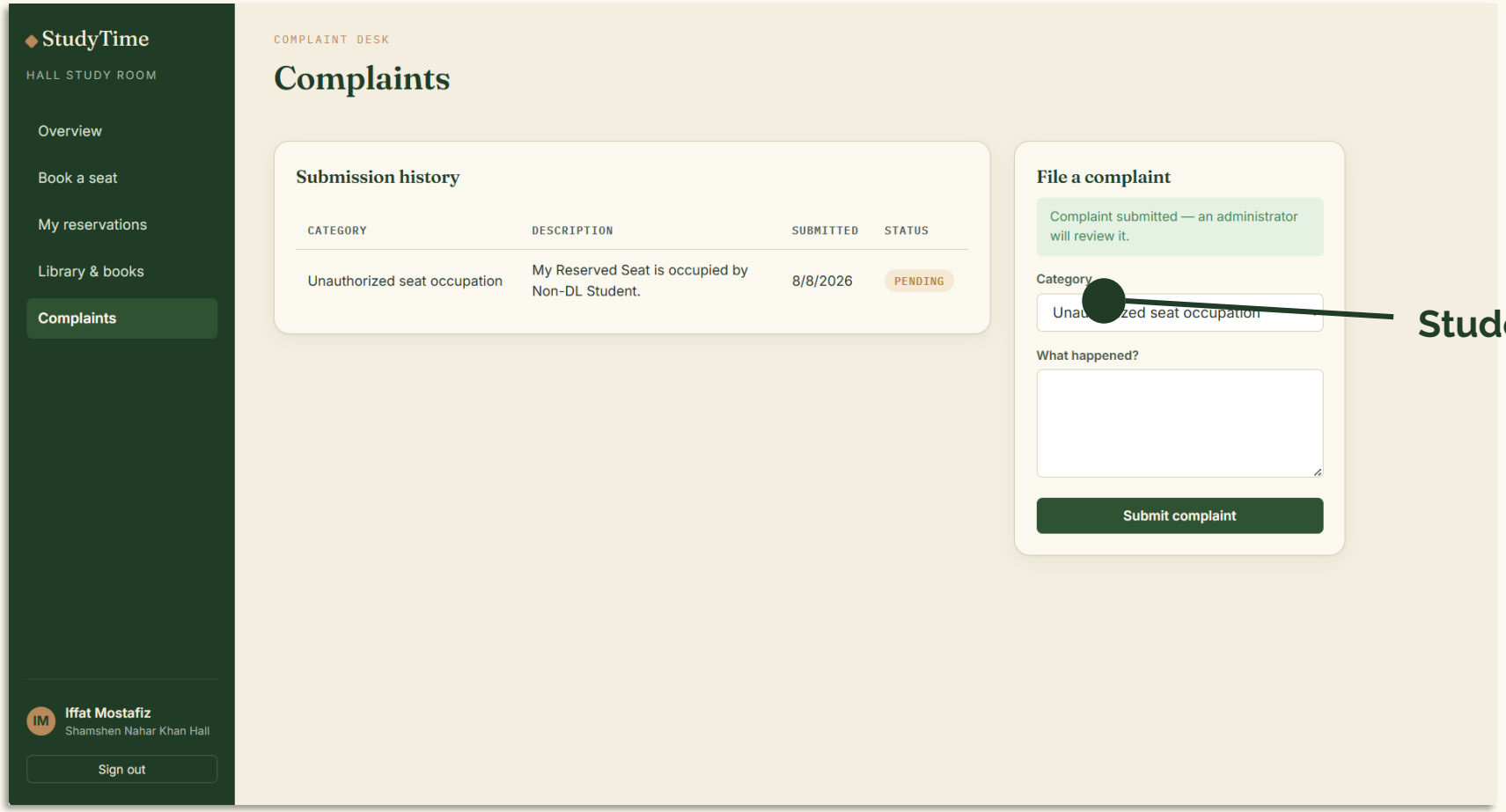
Digital Library Lending & Complaint Management



Digital book catalog with status tracking
(B_Availability)

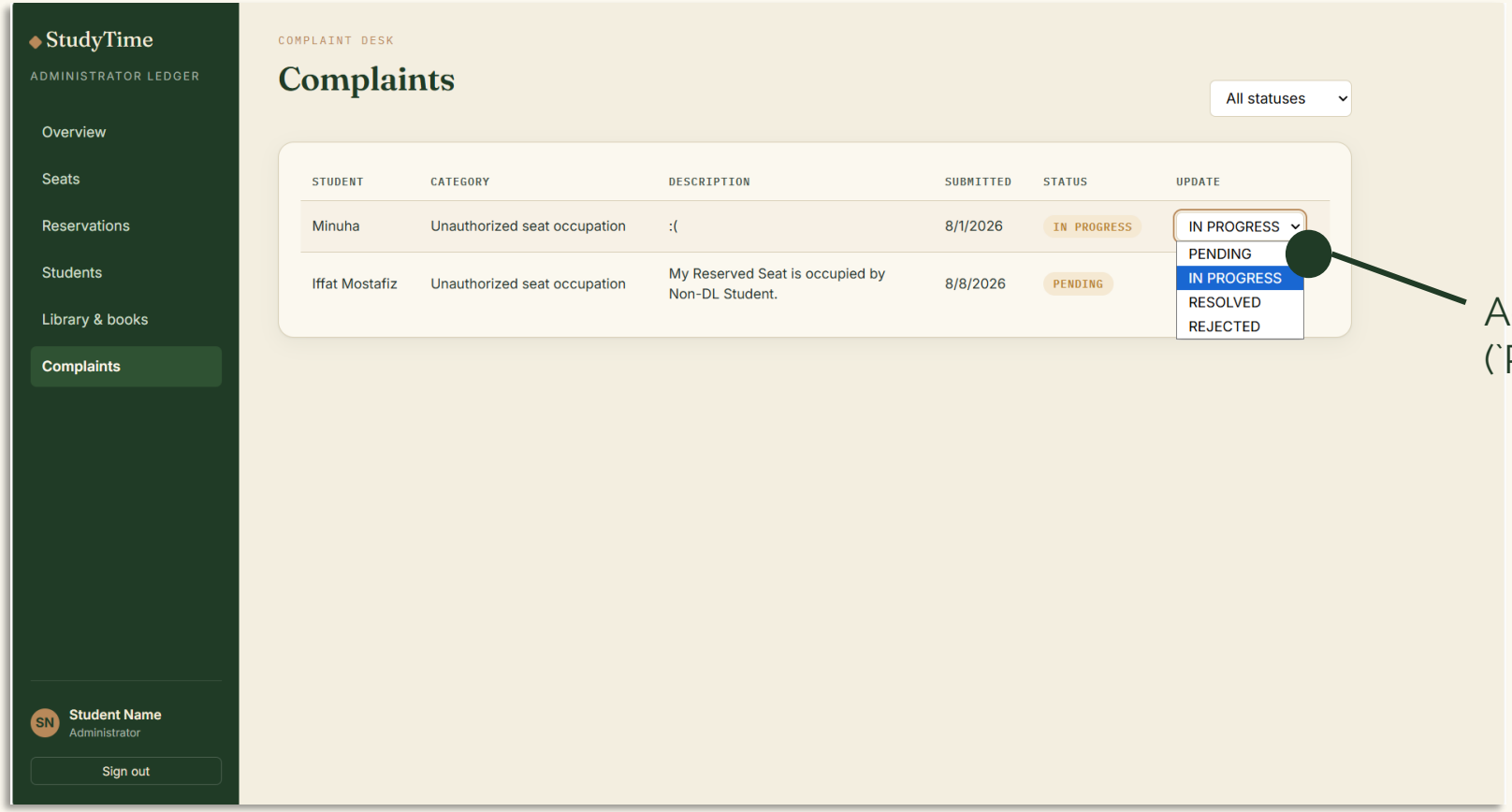
Automated borrow & return date logging
(Borrow_record)

Digital Library Lending & Complaint Management



Students file instant complaints

Digital Library Lending & Complaint Management



Admin dashboard for complaint triage (`Pending` ➔ `In Progress` ➔ `Resolved`).

Full-Stack Technical Architecture

Frontend
React 18 + Vite, Custom CSS Ledger Design System, Interactive Floorplan Component.

Backend
Java 17 + Spring Boot 3, REST Controllers, Spring Data JPA entities.

Database
MySQL 8.0, Auto DDL schema updates, BCrypt password security.



Thank You